

Literature-Implied Answer: Weak-Star Sigma-Discrete Boundaries

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Status

Status: literature-implied negative answer to a boundary subquestion in arXiv:2502.17001.

Source Question

In arXiv:2502.17001, “Polyhedrality for twisted sums with $C(\omega^\alpha)$ ”, Castillo and Salguero Alarcon explain in Section *Boundaries* that a previous conclusion about twisted sums with $c_0(\kappa)$ is uncertain because they do not know whether the existence of a weak-star σ -discrete boundary implies that a Banach space is isomorphically polyhedral.

Supporting Literature

In arXiv:1804.02899, “Topology, isomorphic smoothness and polyhedrality in Banach spaces”, Smith recalls the necessary condition that a polyhedral or LFC norm forces the space to be Asplund and c_0 -saturated. Immediately after this discussion, Smith cites Bible’s thesis, Remark 3.4.4, for a space

$$Z = \ell_1 \oplus \ell_1(S_{\ell_\infty})$$

which is neither Asplund nor c_0 -saturated, and hence is not isomorphically polyhedral, but which nevertheless admits an equivalent norm with a relatively weak-star-discrete boundary.

Identification

A relatively weak-star-discrete boundary is a weak-star σ -discrete boundary, taking the countable union to have one piece. Isomorphic polyhedrality is invariant under equivalent renorming. Therefore Bible’s example, as reported by Smith, gives a negative answer to the implication question:

$$\text{weak-star } \sigma\text{-discrete boundary} \not\Rightarrow \text{isomorphically polyhedral.}$$

Scope

This packet does not settle the full 3-space problem for isomorphic polyhedrality. It only clears the narrower boundary implication used in the source paper to explain why a previous coda was uncertain.

Evidence

The source and supporting PDFs were not locally available in `data/raw/arxiv` at packet creation time. The packet includes the parsed TeX sources `source_2502.17001.tex` and `supporting_source_1804.02899.tex`. The relevant source locations are parsed line 275 of `arXiv:2502.17001` and parsed line 168 of `arXiv:1804.02899`.